

FieldTest Recorder

- **Problem Statement ID** – SIH26231
- **Problem Statement Title** – Digital Companion for Field Drug Testing
- **Theme** – MedTech / BioTech / HealthTech
- **PS Category** – Software
- **Team ID** – 152041
- **Team Name** – DARTH CodersUNI



FieldTest Recorder

A guided, measured and signed field test for NCB's Narcotic Drugs Detection Kit

Proposed solution

- A phone app for NCB's existing Narcotic Drugs Detection Kit. The only addition is a printed colour card. It works offline.
- It guides the officer through Tests A–E in the order of the kit's printed flow charts.
- The officer takes one photo of the reaction, a reagent-only blank and the card.
- The app reads the colour against the kit's colour ranges: positive, no colour change or inconclusive, with every drug on the kit chart that the colour fits.
- Each result is saved as a signed record with time, GPS, officer and photo hash.
- A 14-character code written in the panchnama ties the records to the witnesses' signatures, so anyone can check them later.

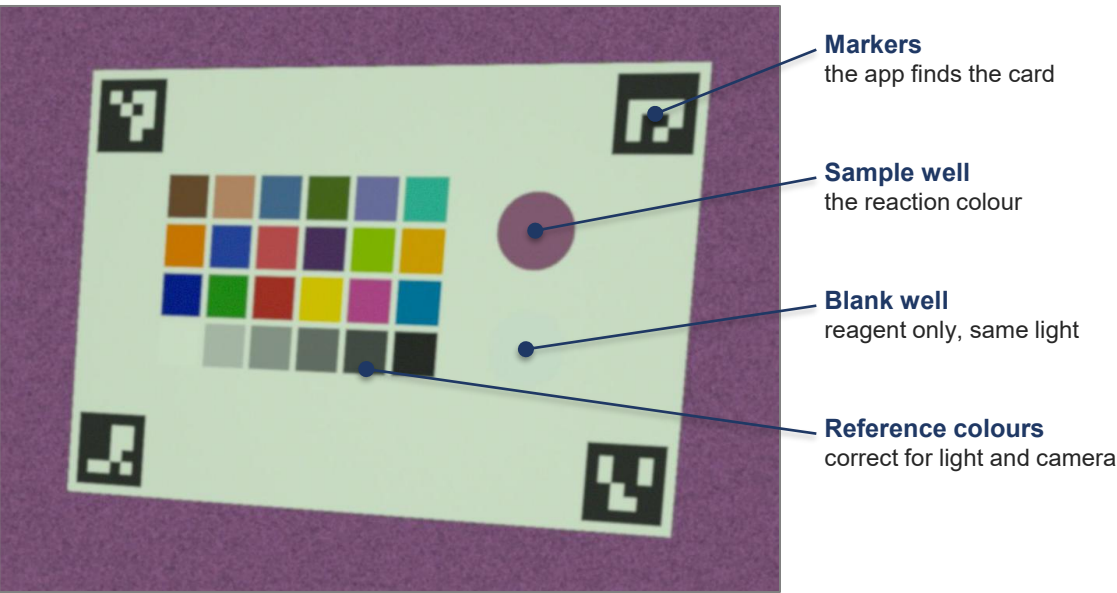
How it addresses the problem

- Today the colour is judged by eye against a printed chart. Here it is measured.
- Bombay HC (2021) called field testing "arbitrary", with no set standard. Here every test leaves a record of when, where and by whom.
- Kerala HC (2024): seized "MDMA" was methamphetamine, a different NDPS quantity category. Our report lists every drug the colour fits and does not name one.

Innovation and uniqueness

- Built on NCB's own kit and flow charts; it is not a generic colour matcher.
- Deleted or repeated tests are detected, not only edited ones (tested in code).
- Still correct when the sample amount varies: 0% false positives in simulation, against 11.7% for a simple rule.

One test in our prototype (simulated photo)

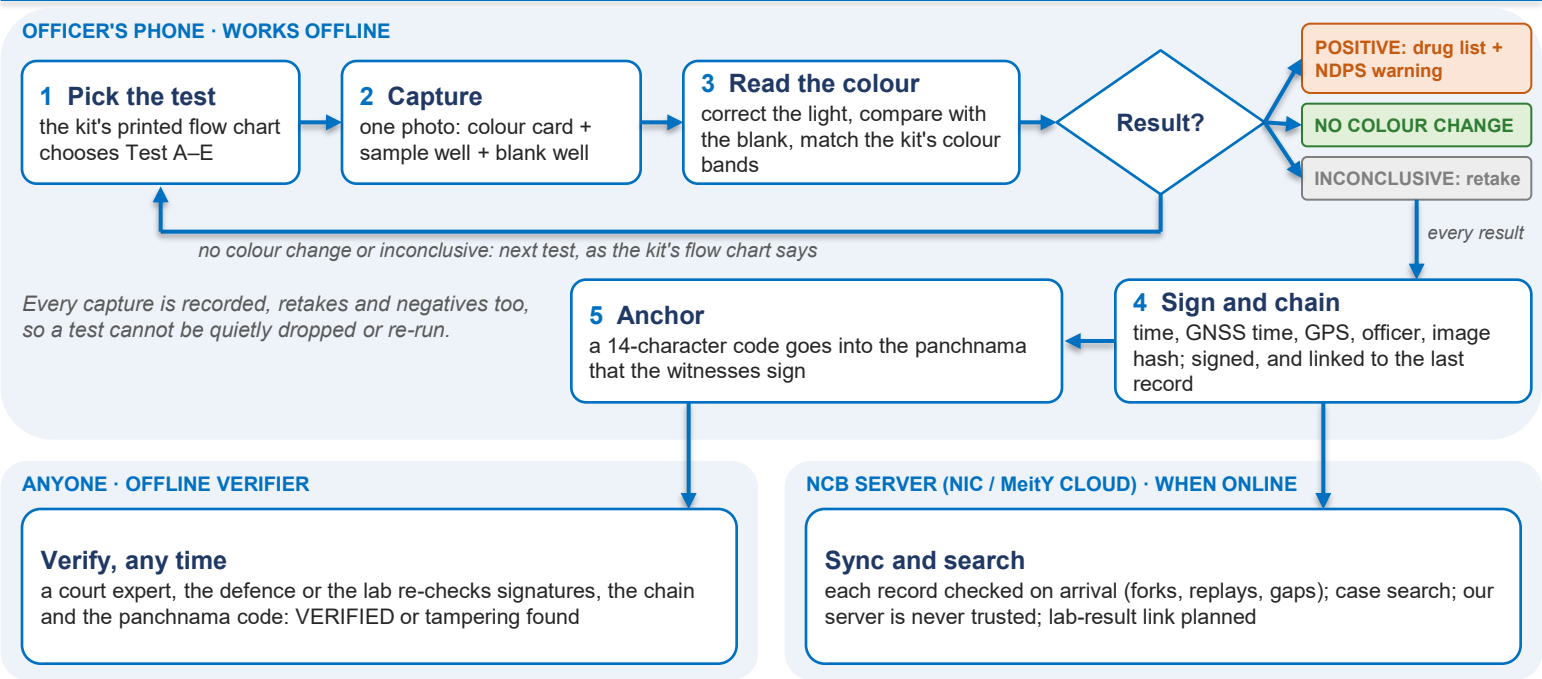


What the signed record for this test says

Test	NCB kit Test A, package P-1
Result	Positive
Colour fits	codeine, heroin, morphine (the kit cannot tell these apart)
NDPS quantity	cannot be decided in the field
Signed with	time, GPS, officer, photo hash; linked to the previous record
Panchnama code	VRBUN-XZCSP-6TMM; the whole raid verifies offline

Output of our Python prototype with test keys. The phone app is the next build.

Process flow: one package, from seizure to court



Every step above runs today in our Python prototype (colour on simulated photos). The Android app and hardware-backed keys are next.

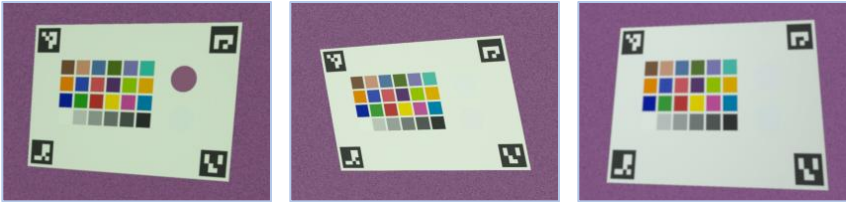
Trust: NCB signs the supervisor list; a supervisor's key (a Class 3 DSC in production) binds officer + phone + key; the phone key is designed to live in StrongBox/TEE (prototype: test keys).

Technologies to be used: Kotlin · CameraX/Camera2 RAW · OpenCV · Android Keystore + attestation · SQLCipher · RFC 8785 JSON · ECDSA P-256 · Spring Boot · PostgreSQL · S3. Prototype today: Python, OpenCV, SQLite.

Deliberately not used: blockchain, cloud AI, a neural-network "drug detector" (a colour test cannot identify a drug; the court must be able to re-run the reading).

Working prototype output

SIMULATED



POSITIVE
Test A: fits codeine, heroin, morphine

NEGATIVE
Test A: no colour change vs blank

INCONCLUSIVE
Test C: kit chart not printed

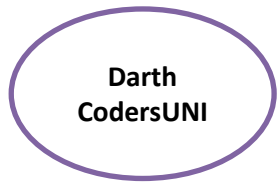
Real reference code on camera-model photos: finds the card, corrects the light, reads the well against the blank, decides, then signs and chains the record; the raid verifies offline.

Implementation status

PROVEN IN CODE Python reference prototype + one-command demo, 65 automated test cases: github.com/Harshith029/fieldtest-recorder

SIMULATED Colour, liveness, night-light and amount experiments on simulated photos

NEXT Android app; printed-card and real-reaction validation with NCB/CFSL chemists



FEASIBILITY AND VIABILITY



Analysis of feasibility: what we have tested so far

What we tested	Result	How
18 ways of tampering with records: edit, delete, hide, forge as server admin	all 18 detected	real code
15 sync failures: lost network, duplicate upload, phone restored from an old backup	all 15 handled correctly	real code
Reading when the sample amount varies	0% false positives (a simple rule gave up to 11.7%)	simulated photos
Sample spilled into the blank well	reported as inconclusive, never as negative	real code
Real phones and real NCB reactions	not tested yet	next: 1,296 tests by NCB/CFSL chemists; target at most 1% false positives

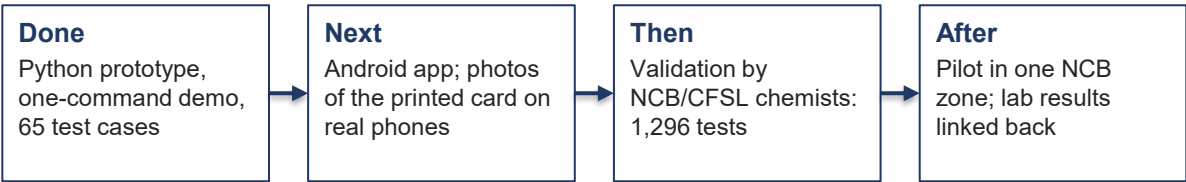
Why it is practical

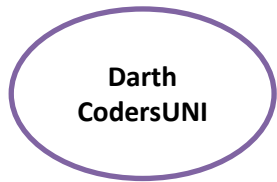
- NCB already supplies the kit free of cost and officers already carry phones. We add a printed card per kit and a few reagent drops for the blank.
- Small infrastructure (estimate): one server pair and 1–3 TB of storage a year on NIC or MeitY-empennelled cloud.
- NCB would hold the signing keys, the colour standard and the code. Open formats, no per-device licence fees.
- No real phone photo or real reaction has been tested yet. That is the next stage, and the table above says how each result was obtained.

Potential challenges and risks, and our strategies

Challenge / risk	Strategy
Real reaction colours differ from the printed chart	Validation by NCB/CFSL chemists under their licences, at 3 sample amounts. We never handle narcotics.
Phones and light vary; glare; night raids	Reference card, RAW capture and a bad-light check (in prototype). The app says "inconclusive" rather than guess. Flash / no-flash night mode is simulated, not yet integrated.
Staged or swapped sample	Planned: capture from before the reagent drop, a liveness check (simulated so far), the package label in frame, a link to the s.105 video.
Rooted phone, insider, deleted records	Supervisor-signed phone binding, panchnama code, server never trusted (in prototype). Hardware keys and attestation are designed; the checker is prototyped.
Access to SIMS / e-Sakshya not confirmed	File exports first. The engine is designed as a module that e-Sakshya could embed.

Path to a working product





IMPACT AND BENEFITS



Potential impact on the target audience

Who	What changes for them
Seizing officer	A guided test and a signed record showing the test was done properly. Planned: Field Test Memo and Form-1 item 5 filled in.
Investigating officer, prosecutor	Records that can be checked, for remand and bail. Planned: a Rule 10(2) grouping check and the lab result linked back.
Courts, defence, the accused	Any expert can re-check the record offline. The quantity category is marked uncertain until the lab confirms.
NCB headquarters and labs	Planned: field result against lab result for every kit batch and test.

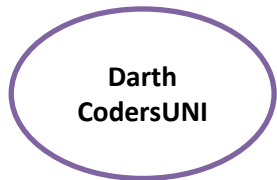
What changes for one seizure

Today (paper)	With FieldTest Recorder
Colour judged by eye	Measured against a signed colour standard
"Dark brown" written in the panchnama	Photo, reading and every drug on the kit chart the colour fits
No proof of time or place	Signed time, GNSS time, GPS and officer
Records can be rewritten later	Deletions and re-runs are detectable
Quantity category guessed	Flagged when it cannot be told in the field
No kit accuracy data	Field result against lab result, per kit batch (planned)

A real case: Anuraj v. State of Kerala (2024). Police alleged commercial-quantity MDMA; the lab found methamphetamine in intermediate quantity (MDMA 0.5 g / 10 g against methamphetamine 2 g / 50 g). A field report must not overstate: ours lists every drug on the kit chart that the colour fits and warns when they fall under different NDPS thresholds.

Benefits of the solution

Social	Economic	Environmental	Scale
Fairer remand and bail decisions. Fewer cases lost on procedure. Honest officers protected.	No new hardware. Reuses the free NCB kit and officers' phones. Small hosting cost.	Paperless records. The only extra consumables are a printed card per kit and a few reagent drops.	1,15,236 NDPS cases in 2022 (MHA). Each seized package can get a recorded field test.



RESEARCH AND REFERENCES



[1] **SIH26231 problem statement** — Narcotics Control Bureau, MHA; sih.gov.in

[2] **NCB Narcotic Drugs Detection Kit** — NICFS, A Forensic Guide for Crime Investigators, ch. 8, Figs 8.12–8.16 (Tests A–E, flow charts)

[3] **NDPS (Seizure, Storage, Sampling and Disposal) Rules 2022** — GSR 899(E): Rules 3, 5, 10(2), 13, 14

[4] **Sagar Parshuram Joshi v. State of Maharashtra** — Bombay HC, 15 Jan 2021 (indiankanoon.org/doc/90970274)

[5] **Masibur Khan v. State** — Delhi HC, 31 May 2023 (indiankanoon.org/doc/91398841)

[6] **Anuraj v. State of Kerala** — Kerala HC, 2024: alleged MDMA found to be methamphetamine

[7] **NDPS small and commercial quantities** — S.O. 1055(E), 19 Oct 2001

[8] **NCB Drug Law Enforcement Field Officers' Handbook** — narcoticsindia.nic.in

[9] **BPR&D SOP: audio-video recording under BNSS; BSA 2023 s.63** — hash value in the electronic-record certificate

[10] **NIJ Standard-0604.01** — Color Test Reagents/Kits for Preliminary Identification of Drugs of Abuse (US DoJ)

[11] **Android key attestation; RFC 8785 (JSON canonicalisation)** — [developer.android.com; rfc-editor.org](http://developer.android.com/rfc-editor.org)

[12] **MHA Rajya Sabha USQ 1832 (11 Dec 2024)** — NDPS case statistics

OUR CODE AND RESEARCH: github.com/Harshith029/fieldtest-recorder

- **Working Python reference prototype with a one-command demo and 65 automated test cases:** record format and offline verifier, colour engine, NCB kit protocol, searchable log, sync protocol. Photos in the demo are simulated; the Android app must match its outputs.
- **17 experiments with 95% confidence intervals**, including 18 tampering attacks and 15 failure scenarios (real code) and colour, sample-amount, night-light and staged-sample tests (simulated photos).
- **Reviewed adversarially, claim by claim:** confirmed defects are either fixed and re-tested or listed openly in the repository's problem register.
- **Safety and law:** our team never handles narcotics; colour tests use published colours, the kit chart and safe dyes. Real-kit validation is designed for NCB or CFSL chemists.